PRISM | Tech Management, SRI-B

CLASH OF THE CLAWS

RVCE_RECALLX

“Think Once, Remember Forever .”

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Problem Statement

Traditional AI assistants and productivity tools fail to retain long-term conversational memory, making interactions repetitive and disconnected. Users often need to re-explain context in every session, reducing efficiency and personalization.

We selected this theme because it aligns with **GenAI + Productivity + On-device intelligent memory systems**, which is a key research direction in next-generation AI assistants.

Who faces this problem:

- Students using AI for study assistance
- Professionals using chatbots for productivity
- Users interacting with AI assistants daily

Why it is important:

- Reduces user efficiency
- Breaks conversational flow
- Limits AI personalization and intelligence

Current Solution & Gaps

Existing AI systems like chatbots or note tools either store data statically or rely on keyword-based search. They lack semantic understanding and cannot retrieve contextually similar past information.

Existing solutions:

- Basic chatbots (Chat GPT-style stateless memory)
- Note-taking apps (manual recall only)
- Simple database-based memory storage

What's missing / broken:

- No semantic understanding of user input
- No long-term memory reasoning
- Poor contextual retrieval
- Repetitive user interactions

Opportunity:

- Build semantic memory-based AI systems
- Improve contextual recall using embeddings
- Create intelligent assistant memory layers

Idea

Your Solutions

We built RECALLX, an intelligent AI memory system that stores and retrieves user interactions using semantic embeddings. It enables context-aware recall instead of keyword matching.

What we built:

- AI-powered memory retrieval system
- Semantic search engine using embeddings
- FastAPI backend with full memory pipeline

Core idea explanation:

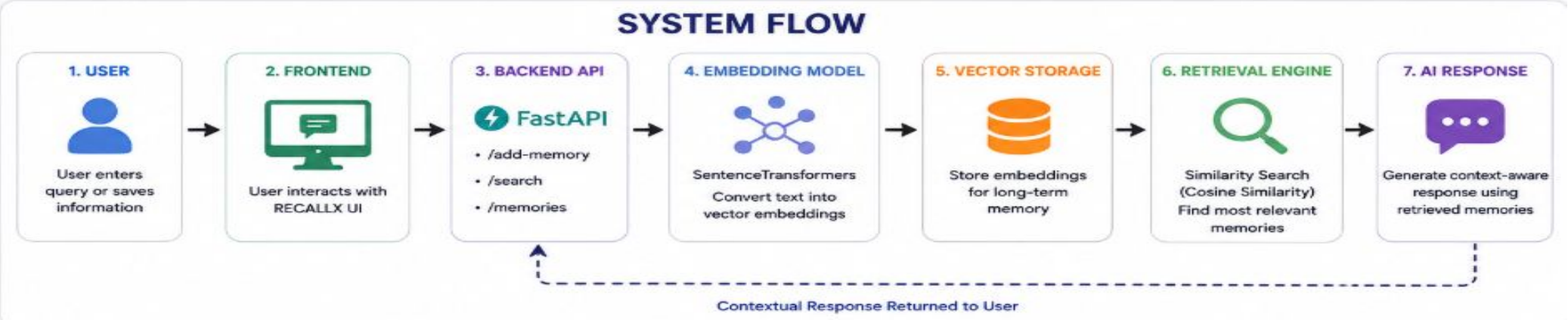
- Convert user text into vector embeddings
- Store them in memory system
- Retrieve similar memories using similarity search

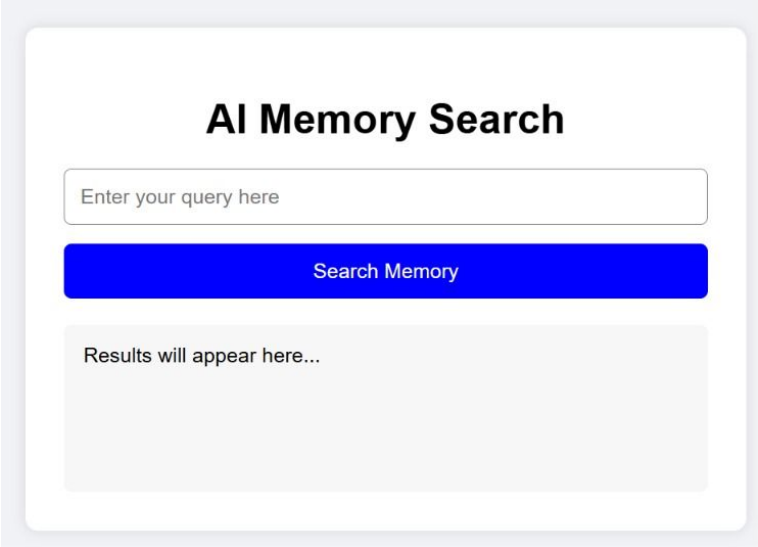
Key features:

- Memory storage system
- Semantic search (not keyword-based)
- Context-aware retrieval
- FastAPI-based API system
- Real-time AI memory interaction

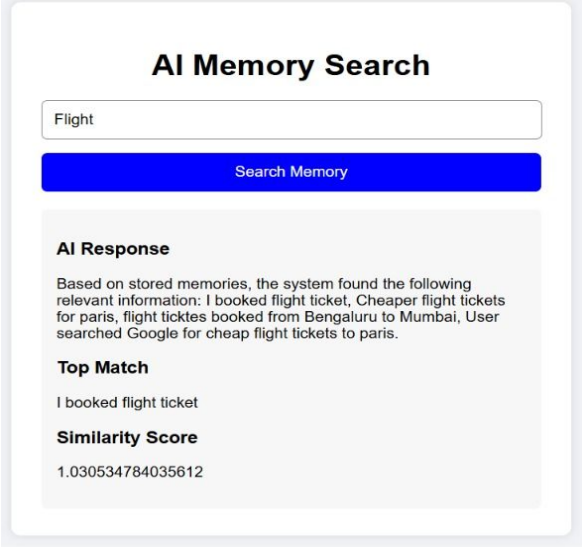
Idea

Demo/Product Walkthrough

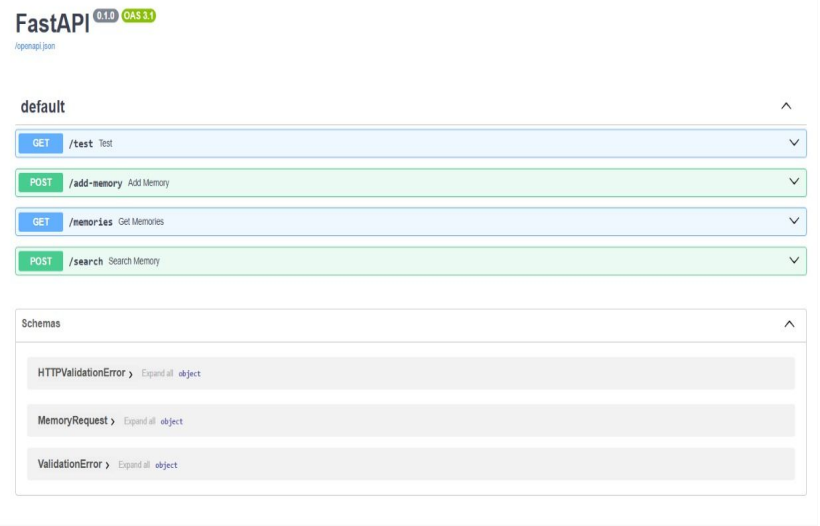




User interface



user interface
(search is found in the memory)



Backend

Tech & Architecture

RECALLX is built using a full-stack AI pipeline combining FastAPI backend with transformer-based embeddings for semantic memory retrieval.

Tech Stack:

Technology	Purpose
Python	Backend Development
FastAPI	API Framework
Sentence Transformers	Embedding Generation
Scikit-learn	Cosine Similarity
NumPy	Vector Operations
JSON	Persistent Storage
Uvicorn	Server

Impact & Use cases

Our system works as a **smart search memory tool** that retrieves previously searched *queries and related text content*, not direct links.

- **Students:** Recall previously searched study topics like IEEE Paper, DSA Project Ideas.
- **Professionals:** Retrieve past searched information such as New Tools, Github login and work-related queries.
- **Developers:** Find earlier searched errors, concepts, and debugging keywords without needing links.
- **General Users:** Rediscover past searches like product to be purchased, Multi Specialist Hospital, or travel queries easily.

User Cases:

User searched these many topics in browser

1. Flight booked from Bangalore to Delhi on June 10
2. Meghana Foods restaurant in Indiranagar
3. Nike shoes from Amazon

Let 1. user query is flight

Result: Flight booked from Bangalore to Delhi on June 10

2. user query is RVCE

Relevant data not found in memory

Idea

Differentiation

Unlike traditional AI systems that rely on exact keyword matching, RECALLX introduces a **semantic memory architecture** capable of understanding meaning, intent, and contextual relationships between conversations. This enables the system to retrieve relevant memories naturally, making interactions more intelligent, personalized, and human-like.

Why RECALLX Stands Out

- Uses semantic understanding instead of basic keyword matching
- Retrieves relevant memories even with different user wording
- Enables long-term, context-aware AI conversations
- Lightweight, scalable, and real-time memory architecture

Core Competitive Advantage (MOAT)

- **Semantic Memory Engine:**
  A reusable AI memory layer that can power any chatbot or assistant
- **Context-Aware Retrieval:**
  Provides meaningful responses based on past interactions, not exact text matches
- **Foundation for Next-Gen AI Agents:**
  Moves beyond stateless chatbots toward memory-enabled intelligent systems

Demo Link

GitHub Repo: https://github.com/RITHIKA0936/-RVCE-_-RECALLX-.git

Demo Link: ▸ Click to Watch Demo Video

Document submitted:

https://drive.google.com/file/d/1ESkNLW3mS_YmSMsSpH80oUjAMVliu_Sk/view?usp